

Reviewer Pack — Minimal Local Indefinite Integral and Communication Complexi...

Entry 03744b3c1396 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Indefinite Integral and Communication Complexity Rank Correlation via Non-Commutative Geometry

Entry ID: 03744b3c1396 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-02 09:57:10 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative Geometry **Field B** (complexity object): Communication Complexity

Statement:

For every boolean circuit C with n inputs, the minimal local indefinite integral (LII) of its associated von Neumann algebra, as defined in non-commutative geometry, is linearly correlated with its communication complexity rank, such that $LII(C) = \Theta(\text{rank_comm}(C))$

Rationale (proposer's reasoning):

Non-commutative geometry provides a framework for studying the structure of quantum systems and their geometric properties. This conjecture aims to explore the potential application of non-commutative geometric invariants to the study of communication complexity, which could expose new structural insights into the computational difficulty of communication tasks.

Taxonomy category: noncommutative_geometry (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 6ffe637aa45b12f9

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For every boolean circuit C with n inputs, if the correlation coefficient between $LII(C)$ and $\text{rank_comm}(C)$ is ≥ 0.8 AND the mean absolute difference between $LII(C)$ and $\text{rank_comm}(C)$ across all seeds is ≤ 3

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 6 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - intitle:noncommutative geometry AND communication complexity AND indefinite integral - title:(minimal local indefinite integral OR LII) AND (communication complexity rank OR rank_comm) - arxiv:math-ph OR arxiv:quant-ph AND (non-commutative geometry OR von Neumann algebra) AND (communication complexity OR circuit complexity)

Top relevant hits considered: - [http://arxiv.org/abs/1209.3595v2] Noncommutative complex differential geometry - [http://arxiv.org/abs/2005.10140v1] On the geometry of null hypersurfaces of indefinite complex contact manifolds - [http://arxiv.org/abs/math/0212372v1] Geometries and Symmetries of Soliton equations and Integrable Elliptic equations - [http://arxiv.org/abs/2206.07611v1] Comment on “Nonperturbative Calculation of Born-Infeld Effects on the Schrodinger Spectrum of the Hydrogen Atom” [Phys. - [http://arxiv.org/abs/0804.3959v2] Nonholonomic systems on Lie algebroids - [http://arxiv.org/abs/0908.0512v2] How hard is it to approximate the Jones polynomial?

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_circuit(n):
        if n == 1:
            return [('NOT', [0])]
        else:
            sub_n = n // 2
            left_circuit = generate_boolean_circuit(sub_n)
            right_circuit = generate_boolean_circuit(sub_n)
            return [('AND', [left_circuit, right_circuit])] + left_circuit + right_circuit

    def communication_complexity_rank(circuit):
        seen = set()
        for gate, inputs in circuit:
            if isinstance(inputs[0], list):
                for sub_input in inputs[0]:
                    seen.add(tuple(gate + [sub_input]))
            else:
                seen.add(tuple(gate + [inputs]))
        return len(seen)
```

```

def minimal_local_indefinite_integral(circuit):
    # Placeholder function, as the actual computation is not defined
    return random.random()

n_values = [5, 10, 15, 20, 30, 40]
metric_values = []
instances_tested = 0
n_max = 0

for n in n_values:
    for _ in range(5):
        circuit = generate_boolean_circuit(n)
        rank_comm = communication_complexity_rank(circuit)
        lii = minimal_local_indefinite_integral(circuit)
        metric_values.append(lii * rank_comm)
        instances_tested += 1
        n_max = max(n, n_max)

correlation_coefficient = sum(metric_values) / len(metric_values)
mean_absolute_difference = sum(abs(x - y) for x, y in zip(metric_values, [x * y for x, y in zip(
    conjecture_holds = correlation_coefficient >= 0.8 and mean_absolute_difference <= 3
    counterexample = "" if conjecture_holds else "mapping_undefined"

    return {
        "metric_name": "Correlation Coefficient",
        "metric_value": correlation_coefficient,
        "instances_tested": instances_tested,
        "n_max": n_max,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19,

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, ...{result}...}}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c95200ce.py", line 78, in <module>
    result = run_trial(seed)
            ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c95200ce.py", line 52, in run_trial
    rank_comm = communication_complexity_rank(circuit)
                ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c95200ce.py", line 35, in communication
    seen.add(tuple(gate + [sub_input]))
               ~~~~~^~~~~~
```

TypeError: can only concatenate str (not "list") to str

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed during execution, which prevents us from verifying the conjecture's conditions. | next: Investigate and fix the error in the test code to allow for a proper verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13636
2	preregistration	ollama_remote	glm4:latest	0	0	9217
3	novelty	ollama_remote	glm4:latest	0	0	10285
4	novelty	ollama_remote	glm4:latest	0	0	9834
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17675
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12480
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7318
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	21475
9	judge	ollama_remote	glm4:latest	0	0	18183

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 120102 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/03744b3c1396.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/03744b3c1396.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/03744b3c1396.tar.gz` (if generated)